

AAKASH SIVASHANMUGAM

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EDUCATION

University of Michigan - Dearborn	MI, USA	GPA: 3.95	2025-Expected Dec 2026
Master of Science in Cybersecurity and Information Assurance			Post Graduate
Vellore Institute of Technology	Bengaluru, India	GPA: 3.92	2024-2025
Cybersecurity Certificate			Post Graduate Program
Sri Sairam Institute of Technology	Chennai, India	GPA: 3.4	2020-2024
Bachelor of Engineering in Computer science and Engineering			Undergraduate

EXPERIENCE

Graduate Research Assistant University of Michigan - Dearborn	Jan 2026 - Present
- Designed a wearable smart ring's architecture by designing a flexible PCB, integrating microcontrollers and sensors, and testing power management and solar charging to safeguard product integrity. - Orchestrated a material-based authentication overlay applied to the solar panel, enhancing security by 20% and paving the way for robust post-prototype feature integration. - Developed a custom Android application to parse mobile sensor data, establishing secure software-to-hardware telemetry and direct interfacing with the embedded device, reducing data latency by 30%.	

Security Automation Intern Danfoss Industries	Apr 2024 - Apr 2025
- Developed automated SOAR workflows using Azure Logic Apps to orchestrate AI agents, playbooks and API integrations automating incident response and triage for Tier 2 SOC analysts reducing manual triage by 40%. - Analyzed EDR telemetry and log sources to enrich alerts with context, improving threat classification accuracy and reducing MTTR by 25%. - Integrated query-based detection rules (KQL/SQL) to identify malicious patterns across Endpoint, Network, and Cloud environments aligning detection logic with MITRE ATT&CK and NIST frameworks.	

Web Developer Sri Sairam Institute of Technology	Mar 2022 - Apr 2024
- Built and maintained a centralized web application for Sairam Institutions, supporting 5,000+ users while incorporating secure application development and data protection practices into student project tracking and management workflows. - Collaborated with executive stakeholders to implement security controls, access management, and organizational security requirements, achieving 100% compliance with organizational security policies while safeguarding sensitive student and project information. - Integrated security considerations into iterative SDLC cycles, delivering four customized progress-monitoring visualizations that achieved a 2x increase in mentor and student engagement while maintaining secure and controlled access to project information.	

PROJECTS

SOC Automation & Log Analysis Lab
- Deployed and optimized a Splunk SIEM in a home SOC lab, integrating 3+ log sources including EDR, firewall, and endpoint telemetry to strengthen centralized threat detection. - Curated 60 validated IOCs from simulated attacks, coded custom SPL detection queries, and performed vendor security reviews through hands-on tool evaluation and gap analysis. - Resolved security events through in-depth PCAP analysis using Wireshark, identifying and documenting malicious network traffic patterns that reduced false positive alerts by 30%.

Nemesis - Advance Deception Engine
- Engineered an active intrusion detection system to monitor and log attacks, simulating production Technology Assurance environments. - Leveraged LLMs (Grok) to automate alert triage, detection-rule generation, and threat intelligence enrichment, reducing analyst cognitive load and improving detection engineering efficiency by 35%. - Conducted security assessments by performing deep log analysis on captured traffic to identify attacker TTPs. - Performed tool security reviews of deception technology, evaluating effectiveness and generating actionable threat intelligence. - Simulated security development lifecycle (SDL) for deception technology, identifying control gaps and driving improvements.

PUBLICATIONS

Phishing Website Detection using Machine Learning
- Developed automated detection logic to flag malicious domains, simulating email security gateway functions for security policy compliance. - Analyzed URL structures to extract IOCs, supporting threat hunting processes aligned with organizational security requirements. - Developed scalable detection logic to enforce security policies and automate threat response, reducing manual analysis time. - Implemented machine learning model achieving 96.6% accuracy, with methodology published in IEEE Xplore digital library.

TECHNICAL SKILLS

Cybersecurity Tools: Nmap, Wireshark, Metasploit, Burp Suite, SIEM(Splunk)
Security Automation: SOAR (Azure Logic Apps), Python/PowerShell Scripting, Automated Triage Workflows
EDR Platform: Microsoft Defender
Security Domains: Identity and Access Management (IAM) Concepts, Technology Assurance, Cybersecurity Operations, Cryptography & Encryption Methods, Endpoint Detection, Cloud Security, EDR Analysis, MITRE ATT&CK
Networking Concepts: TCP/IP, OSI Model, DNS, DHCP, Subnetting
Programming & Scripting Languages: Python, C, C++, Java, JavaScript, Bash, PowerShell, SQL, KQL, SPL
Operating Systems: Linux (Kali, Ubuntu), Windows
Web & Mobile Development: HTML, CSS, JavaScript, Flutter, Dart

CERTIFICATIONS

Fortinet - Fortinet Certified Associate in Cybersecurity, Google - Cybersecurity Professional Certificate, CompTIA - Security+ (In Progress), Cisco Networking Academy - Introduction to Cybersecurity, Cybersecurity Essentials, Introduction to Packet Tracers, Revamp Academy - Full Stack Android App Development, SkillUP - Getting Started with AWS
